WORKED SOLUTION

STEP I 1997, Question 4

Pure • Modulus Equations

STEP I 1997, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s1-q4

The key idea

Each modulus term changes its formula only where its inside is zero, so the four "critical points" $x = -1, 0, 1, 2$ split the real line into five intervals. On each interval every modulus can be replaced by a plain bracket with a definite sign, the left-hand side becomes a linear function, and the equation is easy. The one thing to watch is that a root of the linear equation counts only if it lies in the interval you were working in.

Solution

Write $L(x) = |x+1| - |x| + 3|x-1| - 2|x-2|$. Recall that $|t| = t$ when $t \geq 0$ and $|t| = -t$ when $t < 0$. We work through the intervals from right to left.

$x \geq 2$. All four insides are non-negative, so

$$\begin{aligned} L(x) &= (x+1) - x + 3(x-1) - 2(x-2) \\ &= x + 1 - x + 3x - 3 - 2x + 4 = x + 2. \end{aligned}$$

The equation reads $x + 2 = x + 2$, which is true for *every* $x \geq 2$. So the whole interval $x \geq 2$ consists of solutions.

Ask yourself The equation has become an identity. Have we made an algebra slip?

No. As $x \to \infty$ the coefficients of x in the moduli add up to $1 - 1 + 3 - 2 = 1$ and the constants to $1 - 0 - 3 + 4 = 2$, which exactly match the right-hand side $x + 2$. An equation involving moduli can perfectly well have an interval of solutions, and the question's "find all the solutions" is a hint that something like this might happen. A quick test: at $x = 3$, $L(3) = 4 - 3 + 6 - 2 = 5 = 3 + 2$.

$1 \le x < 2$. Now $x - 2 < 0$, so $|x-2| = -(x-2)$ and

$$L(x) = (x+1) - x + 3(x-1) + 2(x-2) = 5x - 6.$$

Then $5x - 6 = x + 2$ gives $x = 2$, which is not in $[1,2)$. (It is a solution, but we have already counted it.) No new solutions.

$0 \le x < 1$. Now $x - 1 < 0$ as well, so

$$L(x) = (x+1) - x - 3(x-1) + 2(x-2) = -x.$$

Then $-x = x + 2$ gives $x = -1$, not in $[0,1)$. No solutions.

$-1 \le x < 0$. Now $|x| = -x$ too, so

$$L(x) = (x+1) + x - 3(x-1) + 2(x-2) = x.$$

The equation $x = x + 2$ has no solutions at all.

$x < -1$. All four insides are negative, so

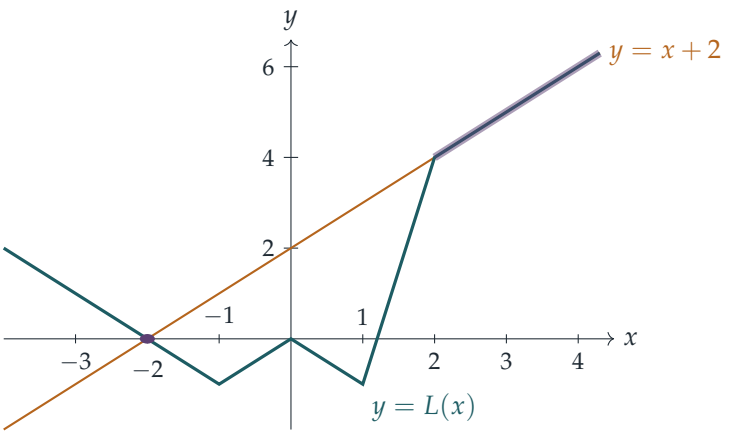
$$L(x) = -(x+1) + x - 3(x-1) + 2(x-2) = -x - 2.$$

Then $-x - 2 = x + 2$ gives $x = -2$, and $-2 < -1$, so this is a genuine solution. Check: $L(-2) = 1 - 2 + 9 - 8 = 0 = -2 + 2$.

Conclusion. The solutions are

$$x = -2 \qquad \text{and every } x \ge 2.$$

The graph shows why: $y = L(x)$ (teal) is a zigzag of line segments, which crosses the line $y = x + 2$ (amber) once at $x = -2$ and then lies along it from $x = 2$ onwards.



Checks

Substituting a few values: $L(-2) = 0 = -2 + 2$, $L(5) = 6 - 5 + 12 - 6 = 7 = 5 + 2$, while at $x = 0$ we get $L(0) = 1 + 3 - 4 = 0 \neq 2$. A computer working in exact fractions confirmed the five linear formulae for L on 100,000 random points, and found that on a grid of step 0.01 over $[-20, 20]$ the equation holds exactly at $x = -2$ and at every grid point with $x \geq 2$, and nowhere else.

Common slips

- Getting $x + 2 = x + 2$ for $x \geq 2$ and writing "no solution" or "infinitely many, so something went wrong", instead of recognising a whole interval of solutions.
- Accepting roots such as $x = -1$ (from $0 \leq x < 1$) that lie outside the interval being considered.
- Forgetting the outer region $x < -1$, which is where the isolated solution $x = -2$ lives.

Beyond STEP

A function like L is *piecewise linear*: continuous, and linear between finitely many "corners". Such functions are central to linear programming and optimisation, where minimising sums of moduli $\sum |x - a_i|$ leads to medians rather than means. They are also a first example of functions that are continuous but not differentiable at isolated points, a theme developed carefully in Abbott, *Understanding Analysis*.

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